

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)	Light travels different distances or by different routes or different paths (1) [Same pulse] takes different times or spreads out over time or equivalent (1)	2			2		
		(ii)	Core only a few wavelengths diameter. <i>Accept thinner <u>core</u>.</i> Accept light can only take one path. Don't accept smaller core.	1			1		
	(b)	(i)	$1.574 \sin C = 1.550 [\sin 90^\circ]$ or equivalent or by implication (1) $C = 80^\circ$ or by implication (1) $\theta = 10^\circ$ ecf on C (1)		3		3	2	
		(ii)	Refraction (1) Accept TIR won't happen Also some reflection or intensity rapidly falls [with successive reflections] or light escapes from the core or light goes into the cladding (1)	2			2		
			Question 8 total	5	3	0	8	2	0